

REPORT

Data Cleaning & Visualization Project

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Data Cleaning & Visualization Project

Objective

The objective of this project is to clean raw data, handle missing values, remove duplicates, and create visualizations to extract meaningful insights from the dataset.

Dataset

Titanic Dataset

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

Missing Values Analysis

- Checked the dataset for missing values using `df.isnull().sum()`.
- Found missing values in Age, Cabin, and Embarked columns.

```
[7]: print(df.isnull().sum())
```

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

Figure 1: Missing Values in Titanic Dataset

Explanation:

This figure shows the number of missing values present in each column of the Titanic dataset before data cleaning. Missing values were identified in the **Age**, **Cabin**, and **Embarked** columns. Identifying missing data is an important step in data preprocessing to ensure accurate analysis and visualization.

Data Cleaning Process

- Filled missing Age values using the mean.
- Filled missing Embarked values using the mode.
- Removed the Cabin column due to excessive missing values.
- Removed duplicate records.

```
[9]: print(df.isnull().sum())
```

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Embarked	0

dtype: int64

Figure 2: Dataset After Handling Missing Values

Explanation:

This figure shows the dataset after the data cleaning process. Missing values in the **Age** column were replaced using the mean value, missing values in the **Embarked** column were filled using the mode, and the **Cabin** column was removed due to a large number of missing entries. This preprocessing step improved the quality and reliability of the dataset for further analysis.

Data Cleaning Steps Performed:

1. Filled missing Age values with the mean age.
2. Filled missing Embarked values with the most frequent value.
3. Removed the Cabin column.
4. Removed duplicate records from the dataset.

Visualizations Created

Survival Count Visualization:

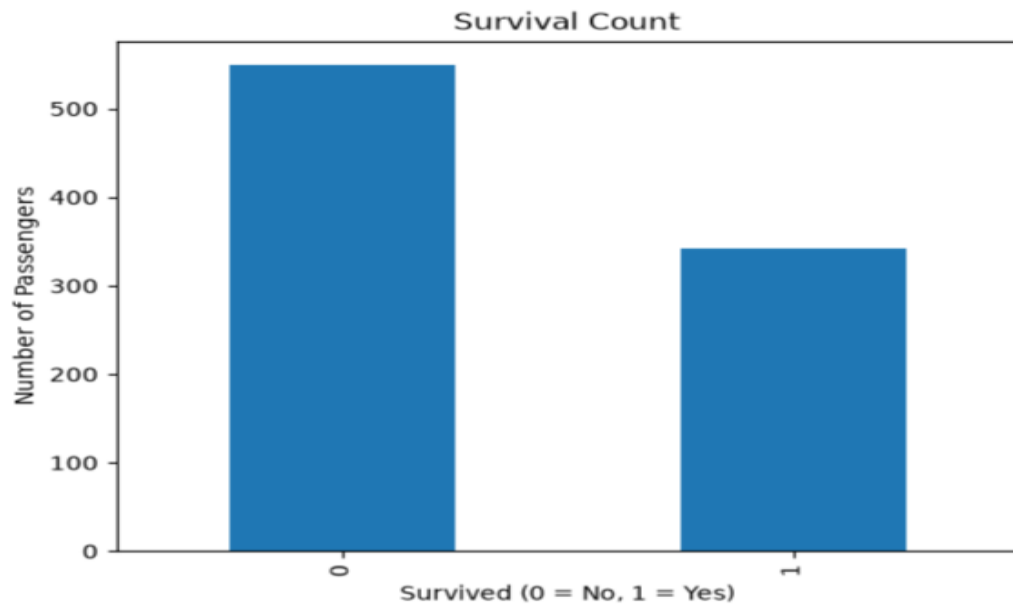


Figure 3: Survival Count of Passengers

Explanation

This bar chart illustrates the number of passengers who survived and did not survive in the Titanic dataset. The chart provides a clear comparison between the two groups and helps in understanding the overall survival distribution of passengers.

Observation

- The number of passengers who **did not survive** was higher than the number of passengers who **survived**.
- This indicates that the survival rate was lower than the mortality rate during the Titanic disaster.

Conclusion

The visualization reveals that a larger proportion of passengers lost their lives compared to those who survived. This insight forms the basis for exploring additional factors such as age, gender, and passenger class that may have influenced survival rates.

Age Distribution Visualization:

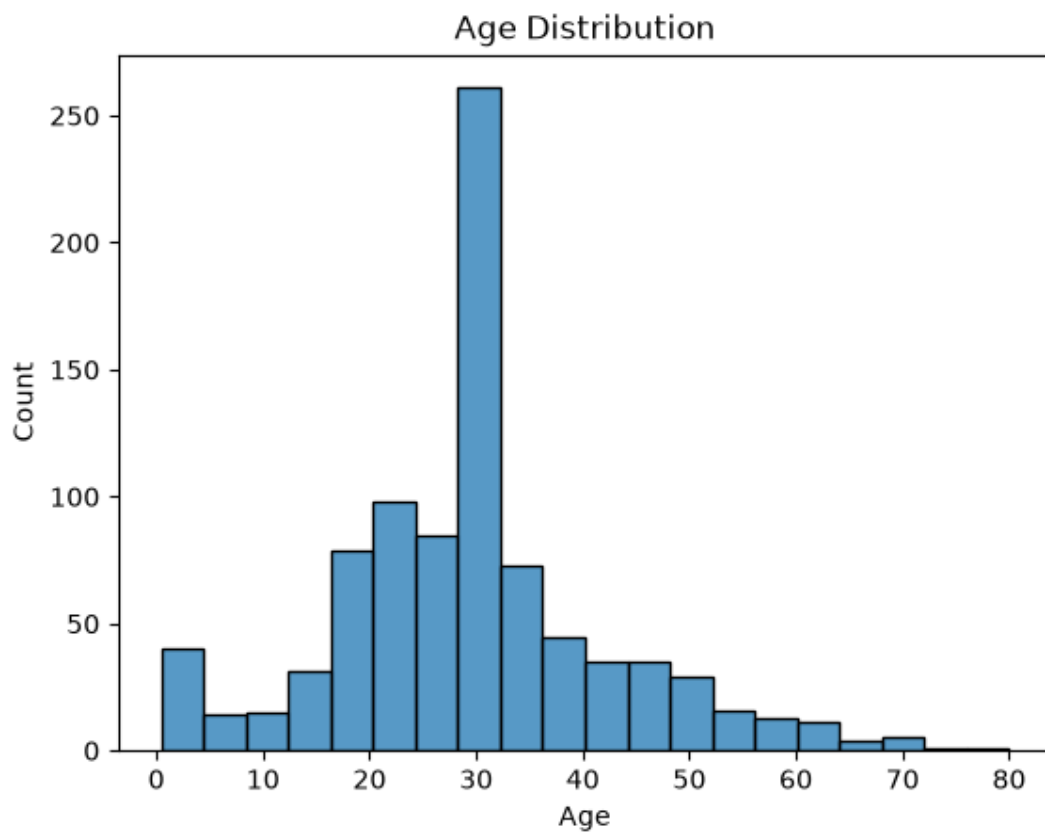


Figure 4: Age Distribution of Passengers

Explanation

This histogram represents the distribution of passenger ages in the Titanic dataset. The chart helps visualize how passengers are spread across different age groups and provides insights into the demographic composition of the dataset.

Conclusion

The analysis shows that the majority of Titanic passengers belonged to the young and middle-aged population. This age distribution provides useful context for further analysis of survival patterns based on demographic factors.

Gender vs Survival Visualization:

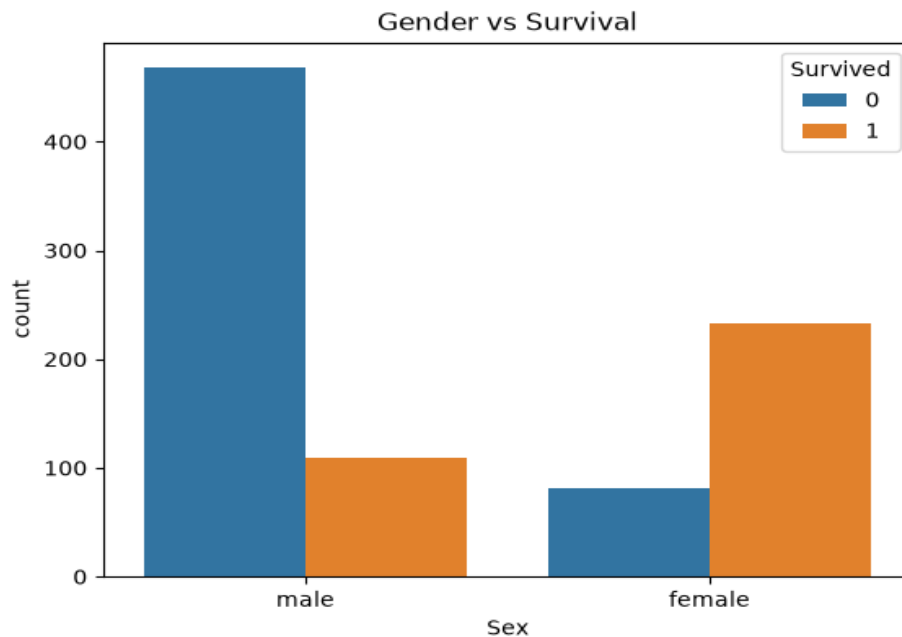


Figure 5: Gender vs Survival Analysis

Explanation

This visualization compares passenger survival status based on gender. The chart displays the number of male and female passengers who survived and did not survive the Titanic disaster.

Observation

- Female passengers had a significantly higher survival rate compared to male passengers.
- A large number of male passengers did not survive.
- The survival count among female passengers was comparatively higher.

Conclusion

The Gender vs Survival analysis indicates that female passengers were more likely to survive than male passengers. This visualization provides valuable insight into one of the key factors influencing passenger survival.

Key Findings and Conclusion

Key Findings

1. Missing values were successfully identified and handled.
2. Duplicate records were removed to improve data quality.
3. Most passengers were between 20 and 40 years old.
4. More passengers did not survive than survived.
5. Female passengers had a higher survival rate than male passengers.

Conclusion

This project demonstrated the process of data cleaning and visualization using Python. Missing values and duplicate records were handled effectively to improve dataset quality. Various visualizations were created to understand passenger demographics and survival patterns. The analysis provided meaningful insights into the Titanic dataset and showcased the practical application of Pandas, Matplotlib, and Seaborn for data analysis and visualization.

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

Outcome

The project successfully achieved its objective of cleaning the dataset and generating visual insights through data visualization techniques.